Unit 3 Review Sheet

Math 9E · Measurement · the essentials on one sheet

3.1 Similar Triangles

Similar	Equal angles + proportional sides · $\Delta ABC \sim \Delta DEF$
Tests	**AA** · **SAS** (2 ratios + included angle) · **SSS** (3 equal ratios)
Thales	DE ∥ BC ⇒ AD/DB = AE/EC · AD/AB = AE/AC = DE/BC
Shadows	$h_1/s_1 = h_2/s_2$ (sun rays parallel)
Scale factor	Length × k · **area × k^2** · **volume × k^3**
Altitude	Right angle at C, CD ⊥ AB: **CD^2 = AD·DB** · AC^2 = AD·AB
Pythagoras	$a^2 + b^2 = c^2$

★ Trigonometry

SOH CAH TOA	sin = opp/hyp · cos = adj/hyp · tan = opp/adj
Special	sin 30° = ½ · cos 30° = √3/2 · tan 45° = 1
Equilateral	R = a/√3 · area = $(\sqrt{3}/4)a^2$ · height = $(\sqrt{3}/2)a$

3.2 Unit Conversions

Method	value × (to ÷ from)
Prefixes	T 10^{12} · G 10^9 · M 10^6 · k 10^3 · c 10^{-2} · m 10^{-3} · µ 10^{-6} · n 10^{-9} · f 10^{-15}
Imperial	1 in = 2.54 cm · 1 ft = 12 in · 1 mi = 5280 ft = 1.609 km
Land	1 ha = 10 000 m^2 = 2.47 acres · 1 acre = 4840 yd^2
Area / volume	**Square / cube the factor**: 1 m^2 = 10^4 cm^2 · 1 m^3 = 10^6 cm^3
Dose	(ordered ÷ on hand) × quantity · 1 g = 1000 mg

3.2 Surface Area

Cylinder	$2\pi r^2 + 2\pi rh = 2\pi r(r + h)$
Cone	$\pi rs + \pi r^2 = \pi r(r + s)$ · **$s = \sqrt{(h^2 + r^2)}$**
Sphere	$4\pi r^2$ · hemisphere (solid) $3\pi r^2$
Square pyramid	$b^2 + 2bs$ · **$s = \sqrt{(h^2 + (b/2)^2)}$**
Box	2(lw + lh + wh)
Composite	Outside faces only · drill a hole: − openings + inside wall

3.3 Volume & Capacity

Prism · cylinder	$V = Bh$ · $V = \pi r^2 h$
Pyramid · cone	$V = \frac{1}{3}Bh$ · $V = \frac{1}{3}\pi r^2 h$
Sphere · hemisphere	$V = (4/3)\pi r^3$ · $(2/3)\pi r^3$
Capacity	$1\ cm^3 = 1\ mL$ · $1\ m^3 = 1000\ L = 1\ kL$ · $1\ gal = 3.785\ L$
Cut pyramid / cone	$V = V_{big} \times (1 - (d/H)^3)$, d from apex
Water in cone	$V_{water} = V_{tank} \times (depth/H)^3$

3.3 Cavalieri's Principle

2D rule	Same height + **same width** at every level ⇒ **same area**
3D rule	Same height + **same cross-section area** at every level ⇒ **same volume**
Key idea	The **area** of each slice must match, not its shape (Carl vs Hiva: triangle 30 cm^2 = rectangle 30 cm^2)
Oblique solids	Slanted prism or cylinder = right one with the same base and height: $V = Bh$
Sphere proof	Slice at height h: hemisphere $\pi(r^2 - h^2)$ = washer $\pi r^2 - \pi h^2$ ⇒ hemisphere = cylinder − cone = $\frac{2}{3}\pi r^3$ ⇒ sphere = $(4/3)\pi r^3$
Pyramid / cone	Same base area and height ⇒ same volume, whatever the base shape: $V = \frac{1}{3}Bh$

3.3 Solve for r, h, s

Cylinder	$h = V/(\pi r^2)$ · $h = (SA - 2\pi r^2)/(2\pi r)$
Cone	$r = \sqrt{(3V/(\pi h))}$ · $r = L/(\pi s)$ · $h = \sqrt{(s^2 - r^2)}$
Sphere	$r = \sqrt[3]{(3V/(4\pi))}$ · $r = \sqrt{(SA/(4\pi))}$
Pyramid	$h = 3V/B$ · square base $b = \sqrt{(3V/h)}$
Similar solids	$k = \sqrt[3]{(V_2/V_1)} = \sqrt{(A_2/A_1)}$

3.4 Scale

Scale	model : actual, same units
Types	Reduction 1 : n · enlargement n : 1
Scale factor	bigger ÷ smaller · **× for bigger, ÷ for smaller**
Map area / model volume	$\times n^2$ · $\times n^3$

⚡ Shortcuts & Traps

3 : 2 : 1	Cylinder : sphere : cone (same r, h = 2r)
Triples	3-4-5 · 5-12-13 · 8-15-17 · 7-24-25
Half-depth cone	Holds only ⅛ of the volume
Doubling	Lengths × 2 ⇒ area × 4, volume × 8
Quick math	km/h ÷ 3.6 = m/s · tax 12% ⇒ × 1.12 · check with $\pi \approx 3$
Watch out	Diameter → radius · slant vs height · round buckets **up** · base included?